🧪 QUESTIONS

1️⃣ DATA LOADING & UNDERSTANDING

1. After loading the **`supermarket_data.csv`** file, list three key columns from the dataset and state whether each is **numerical or categorical**.
2. Why is it important to inspect the dataset before performing statistical analysis?

2️⃣ DESCRIPTIVE STATISTICS (Item Price)

3. Using the *item price* data, calculate and record:
   - Mean
   - Median
   - Variance
   - Standard Deviation
4. Interpret your results:
   - If the mean and median are close, what does it say about price distribution?
   - What does a high standard deviation indicate about supermarket pricing?

3️⃣ PROBABILITY DISTRIBUTIONS (Based on Dataset Columns)

5. Using the **customer salary** column, explain whether the salary distribution appears normal (bell-shaped) using its mean and standard deviation.
6. Calculate the probability of selecting a **female customer** from the dataset. What business insight can be drawn from this probability?
7. The supermarket offers random discounts. Explain why this scenario can be modeled using a **uniform distribution**.

4️⃣ INFERENTIAL STATISTICS (Sampling & Confidence Interval)

8. A random sample of 50 customer totals was taken. Using this sample:
   - What is the sample mean of total spending?
   - Construct a **95% Confidence Interval** for true average spending.
9. Interpret the Confidence Interval in plain language, such as:
   _“We are 95% confident that the true average spending is between ___________ and __________.”

5️⃣ HYPOTHESIS TESTING (t-Test: Male vs Female Spending)

10. State the **Null Hypothesis (H_0)** and **Alternative Hypothesis (H_1)** when comparing total spending between **male and female** customers.

11. Based on the p-value obtained, decide:

- Is there a significant difference in spending between genders?
- What marketing decision could the supermarket make from this result?

6️⃣ CHI-SQUARE TEST (Gender vs Product Preference)

12. What does the Chi-Square test evaluate in the context of gender and product preference?

13. If the p-value is below 0.05, what does this indicate about the relationship between gender and product choices?

14. How can this result help supermarket stocking and promotions?